

CHARLES BENELLO

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EDUCATION

The University of Chicago | GPA: 3.7/4.0

Chicago, IL

Master of Science in Financial Mathematics

Expected Mar 2027

- Coursework: C++ in Finance, Distributed Systems, Reinforcement Learning, Probability & Stochastic Processes, Options Pricing, ML in Finance

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean

Jun 2025

- Coursework: Algorithms (Honors), Data Structures, Operating Systems, Computer Architecture, Concurrency Control, Databases, Networks

TECHNICAL SKILLS

Languages: Python, C++, Rust, Go (Raft, gRPC), SQL, CUDA, TypeScript

Frameworks & Tools: NumPy/Pandas, PyTorch, SciPy, scikit-learn, Polars, Arrow/Parquet, Git, Docker, CMake, Linux, PostgreSQL, DuckDB, SLURM

AI Tooling: Claude Code, Codex

EXPERIENCE

Software Development Engineer Intern (Incoming)

Aug – Nov 2026

Amazon Web Services, Redshift Query Optimizer (planner) team: data-lake query performance

Palo Alto, CA

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- Designed CrocSort (VLDB 2026): a skew-resilient parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort, benchmarked across skewed and uniform key distributions
- Sped up random access 3× and cut memory 40% by engineering Small String Optimization across storage structures, inlining short keys into nodes to eliminate heap allocations on the hot path
- Authored a second paper (under review) on a lightweight model that predicts optimal sort configuration on cloud and disaggregated hardware, transferring calibration across machines to within 3–5% of optimal

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab)

Lausanne, Switzerland

- Accelerated out-of-memory analytical queries 1.5–3× by engineering a macroblock-based GPU decompression strategy in CUDA/C++, streaming compressed macroblocks through device memory and keeping decompression on-GPU instead of round-tripping through host RAM
- Drove a new cost-based compression research direction by benchmarking GPU compression under resource-contention models, enabling adaptive decompression choices that trade decompression cost against bandwidth and contention across TPC-DS workloads

Machine Learning Engineer Intern

Mar – Jun 2026

PayPal *Chicago, IL*

- Trained a self-supervised transaction foundation model (TabBERT-style hierarchical transformer, ~5M params) on 14.5M unlabeled transactions, reaching AUC 0.974 / F1 0.822 on downstream fraud detection at a 0.12% positive rate
- Ran 87 ablations on a SLURM cluster across masking and pretraining objectives; per-field masking alone added +1.9% AUC and +16.8% F1

Machine Learning Engineer Intern

Oct – Dec 2025

Bank of America

Chicago, IL

- Built a portfolio-rebalancing platform that converts capital and liquidity constraints into risk-minimizing trade recommendations, stress-tested across probability-weighted macro scenarios

Software Engineering Intern

Mar – Aug 2026

Alter Domus

Chicago, IL

- Lifted document-extraction accuracy from a 50% baseline to 92–97% across document types by architecting a multi-stage LLM pipeline with schema-enforced structured output and cross-reference verification on financial filings over 100 pages

Teaching Assistant

Sep 2022 – Mar 2026

University of Chicago, Computer Science & Mathematics

Chicago, IL

- Awarded the 2026 Wayne Booth Graduate Student Teaching Prize, UChicago's top graduate teaching award
- TA'd 11 course-sections across computer science and mathematics: Head TA for Database Systems (3×), Systems Programming I–II, Intro to CS I–II, Calculus I, and Intro to Proofs in Analysis (3×)

PROJECTS

GPU External Sort at Terabyte Scale | CUDA/C++ | github.com/cmbenello/gpu-research

2026

- Sorted 1.08TB (9B records) in ~5 minutes on a single H100 by streaming NVMe data directly into GPU memory with GPUDirect Storage; up to 20.1× faster than DuckDB and 34.5× faster than Polars, which fail beyond ~200GB, making it the only tested system that completes at terabyte scale

Relational Database Engine | Rust | github.com/cmbenello/CrustyDB

2023

- Implemented a multi-threaded relational database from scratch: B+ tree indexing, a concurrent buffer pool with page-level locking, heap storage, and a SQL query-execution engine supporting projection, filters, joins, and aggregation

Transit Coverage Optimizer | Python | cmbenello.github.io/covtsp

2026

- Computed a route visiting every station on the London Underground 60 minutes faster than the standing Guinness World Record (16h45m vs. 17h45m) by solving the Covering-TSP over a 160K-node / 1.26M-edge time-expanded graph; generalized to NYC and Berlin with an interactive demo

PUBLICATIONS

[1] R. Otaki*, C. Benello*, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)

[2] C. Benello, R. Otaki, et al. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." (Under review).

[3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

ADDITIONAL

Citizenship: US, UK, and Italy | **Interests:** Prediction markets, Rainbow Six Siege (peak top 0.1% of ~2M ranked players), weightlifting, running (3× Chicago Marathon), climbing